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Repeated Games:

1 \ 2	C	DC
C	2, 2	-1, 4
DC	4, -1	0, 0

$\Longrightarrow$ Equilibrium {DC, DC}
DC dominant

Care more about early rounds
↳ $x + x\delta + x\delta^2 + \dots$

- common discount factor $:= \delta$
- stage game is game played each round

- strategy in round 2 is function mapping outcome in R1 to strategy in R2
- i.e.: game played twice. round 1 strategy $\in$ set of all dist. over C, DC. round 2 strategy $\in$ set of all functions f: outcome in round 1 $\to$ dist over C, DC

- $H^t :=$ history up until time t; what are outcomes of rounds until t.

Suppose Prisoner's Dilemma is played 10x
- equilibrium comes to {DC, DC}. in R10, play DC, DC so each round is (DC, DC) because final strategy known. case for any finite amt of rounds.

Finiteness allows for backwards induction, Start at end, find SPNE and go back.

Grim Trigger Strategy:
- Play C in R1, cooperate in t if C, C is outcome of all prior periods, play DC otherwise.
  - ⤷ can be augmented for other equilibria.

Only need to check for one-shot deviation principle:
- do different action from strategy in some t, then return to strategy in $t+1$. if profitable, deviates, else passes check.

Period T: (Holding 2 fixed)
- case 1: {C, C} played in all t before t; payoff $\frac{2}{1-\delta}$ if followed
  - ⤷ deviate: {DC, C} played in t; payoff of 4 in t + 0 after bc {DC, DC} will be played.
  - $\frac{2}{1-\delta} \geq 4 \Rightarrow \delta \geq \frac{1}{2}$
- case 2: {C, C} wasn't played in all t before t. {C, DC} $\Rightarrow -1 \Rightarrow$ {DC, DC} $\Rightarrow 0 \Rightarrow \frac{2}{1-\delta} \geq -1$

To practice, change numbers and repeat. Try w/ Bertrand competition + Cournot.

Rubenstein Bargaining Game:

- divide pie b/w 2. P1 first, P2 accept or reject. P2 goes, P1 accept/reject ... repeating forever

$\delta_i :=$ player i's discount factor

2 round:
P2: $0 + 1 \cdot \delta_2 = \delta_2$ $\therefore$ P1 should propose $(1-\delta_2, \delta_2)$ as split. P2 strategy is A if offered $\geq \delta_1$ in R1, reject otherwise to get δ_2 in R2, proposing $(0,1)$ w/ P1 accepting any offer.

- outcome is P2 accepts. needs to be $\geq$; $>$ leads to loss of equilibrium.

take limit to find inf. solution.

Player 1's proposal in inf: $\frac{1-\delta_2}{1-\delta_1\delta_2}$ (what they keep)
Player 2 gets $1 - \frac{1-\delta_2}{1-\delta_1\delta_2}$

Rubenstein in Econometrica.
↳ Nash too

Exam Comments: